DAWOOD UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF BASIC SCIENCES AND HUMANITIES

Subject:	Applied Physics	
Course code:	BS-1001	
Discipline:	Bachelor of Cyber Security	
Semester:	First Semester, First year	
Effectiveness:	Batch-2023	
Course type:	Compulsory	
Marks:	Theory: 100	Practical: 50
Credit Hours:	3 CH	1CH
Teaching Scheme:	3HRS/week	3HRS/week
Assessment:	20% sessional, 30% Mid semester, 50% Final Exam	

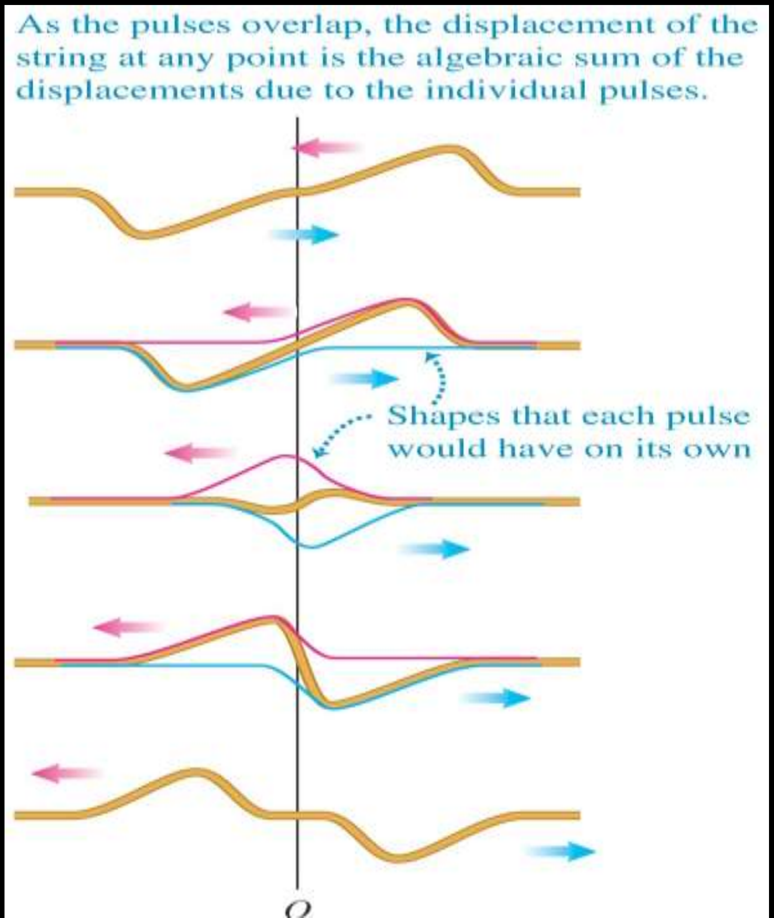
Books:

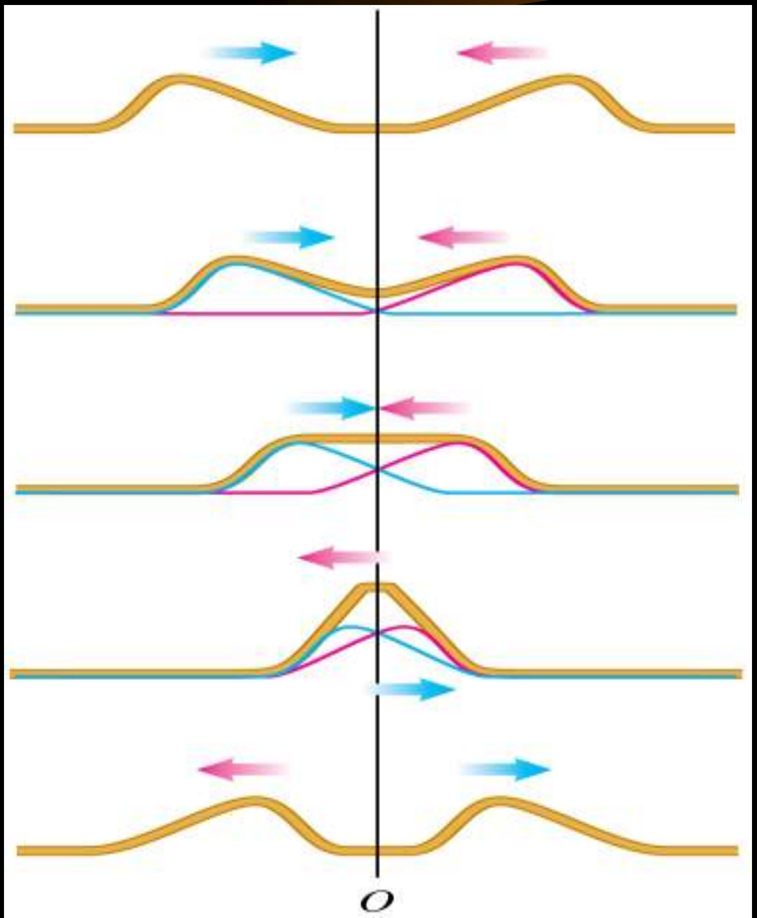
1. physics For scientists and engineers by Serway Jewett
2. Semi conductor Physics by Macalve
3. University physics by Freemen and Young

Waves interference

Wave interference and superposition

- *Interference* is the result of overlapping waves.
- *Principle of super-position*: When two or more waves overlap, the total displacement is the sum of the displacements of the individual waves. Study Figures at the right.





Superposition

- Where the two water waves meet, the motion of the water is a sum, a **superposition**, of the waves.

- You'll learn how this **interference** can be constructive or destructive, leading to larger or smaller amplitudes.

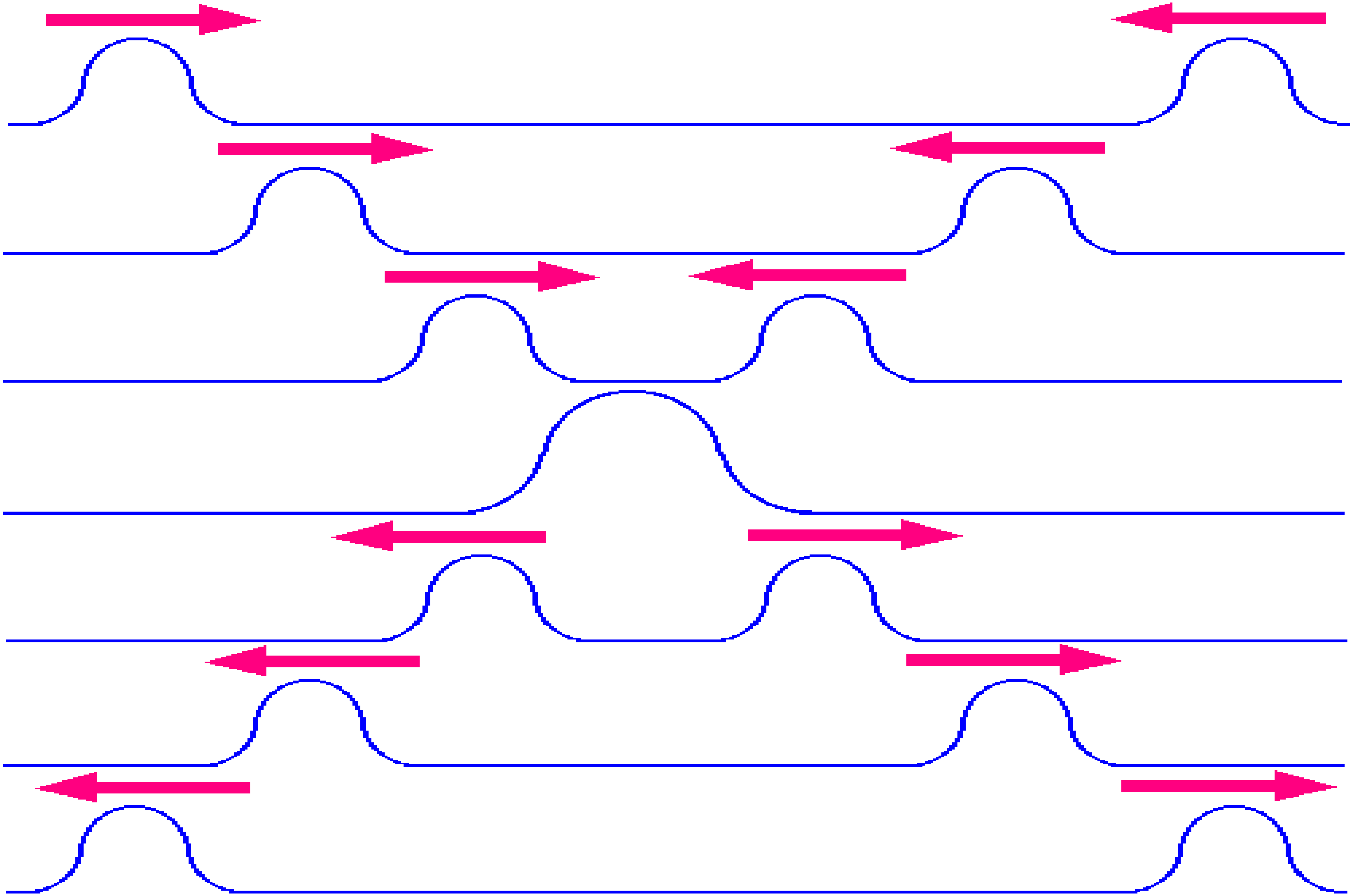
Wave Interference

- When we had two particles (carts) and pushed them into each other they collided and then stopped.
- When waves collide they temporarily interfere with one another, but they do not stop each other.

Wave Interference

- The Principle of Superposition states that the displacement of a medium caused by two or more waves is the algebraic sum of the displacements caused by the individual waves.

Superposition (Interference)



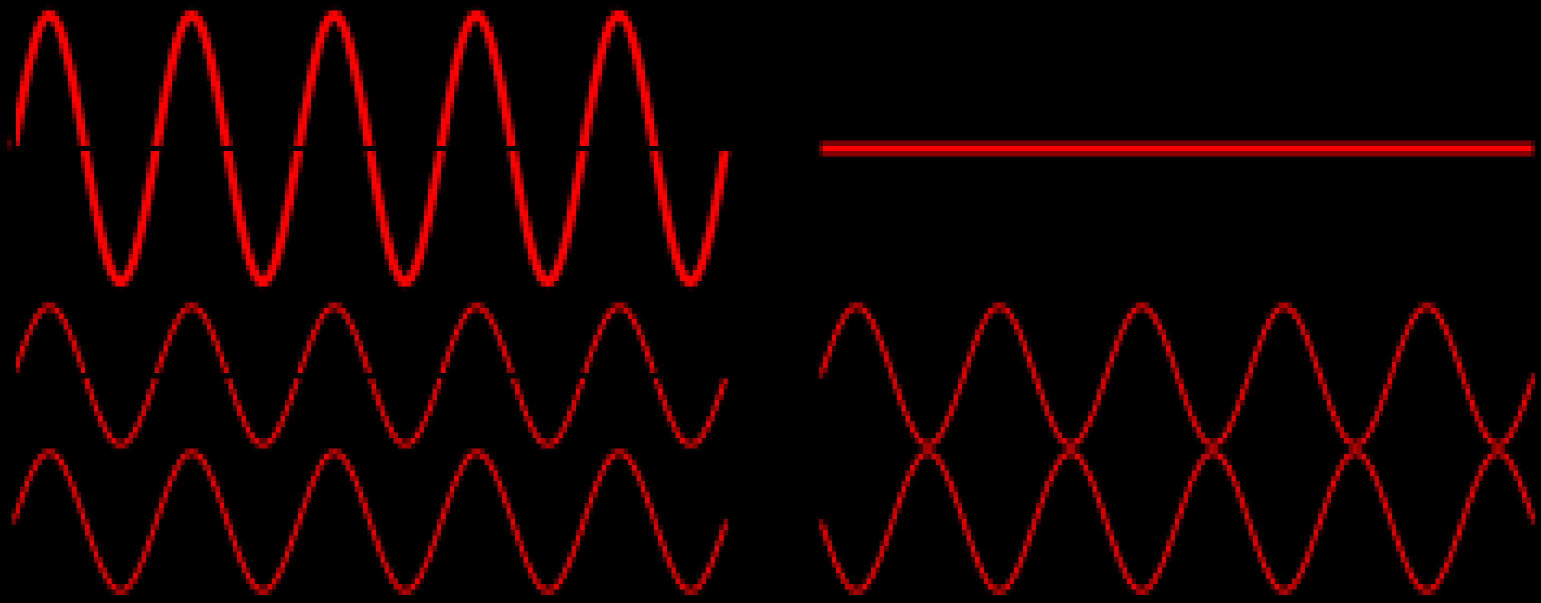
Types of Interference

- There are three types of interference

1. <u>Constructive Interference</u>: When the crest of a wave meets a crest of another wave of the same frequency at the same point.
2. <u>Destructive Interference</u>: When the crest of a wave meets a trough of another wave of the same frequency at the same point.
3. <u>Different Amplitude</u>: All other scenarios of waves interference.

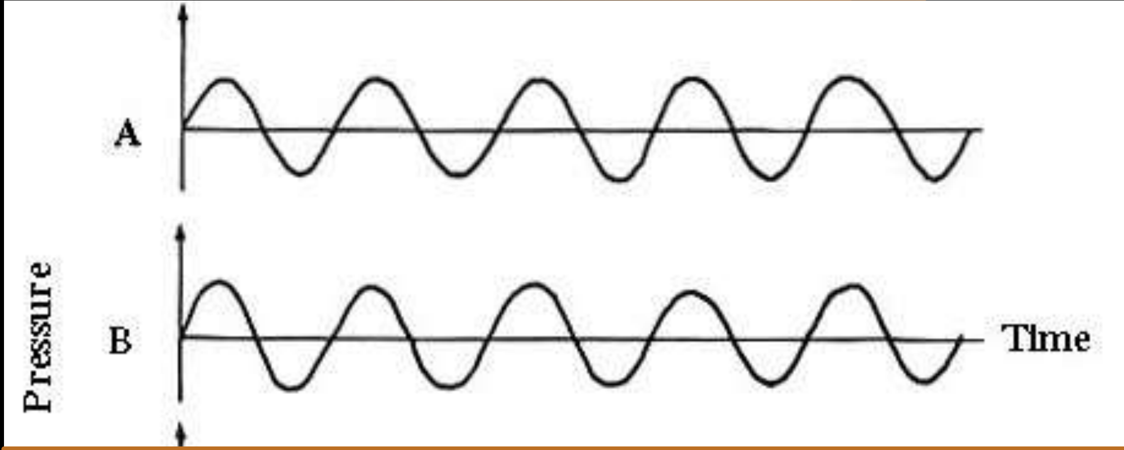
Constructive vs. Destructive

- Which is constructive and which is destructive interference? How do you know?



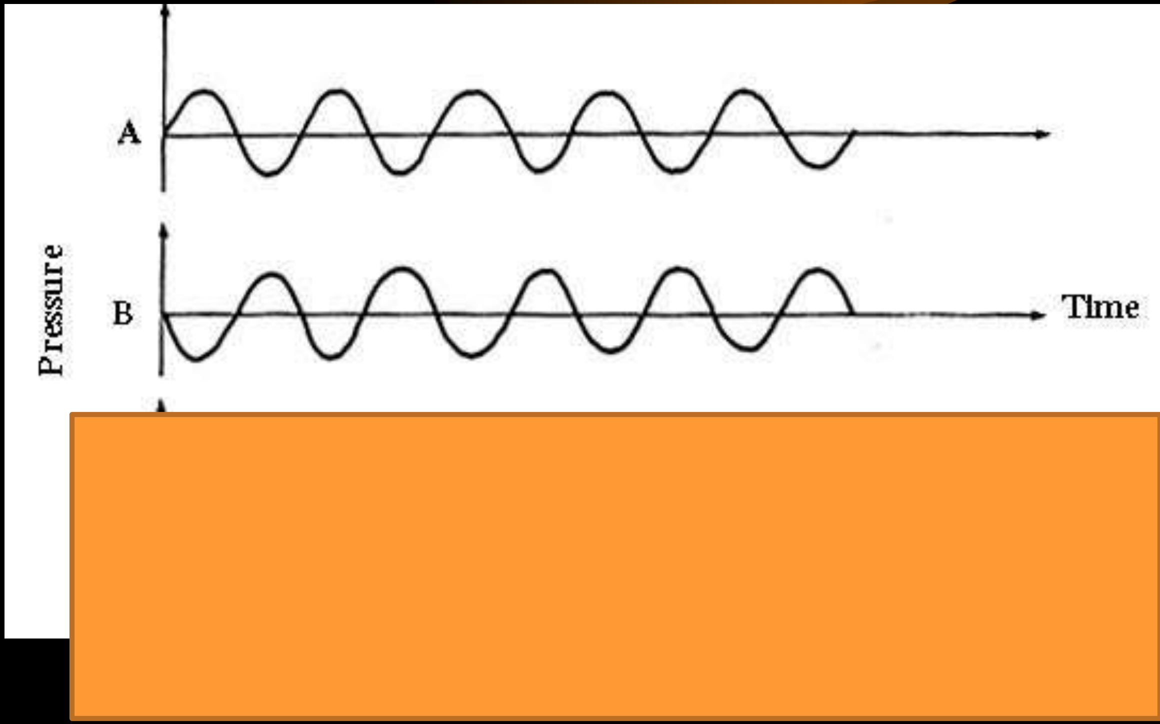
Interference Practice

- If waves A and B were aligned on top of each other, what would the resultant wave look like?



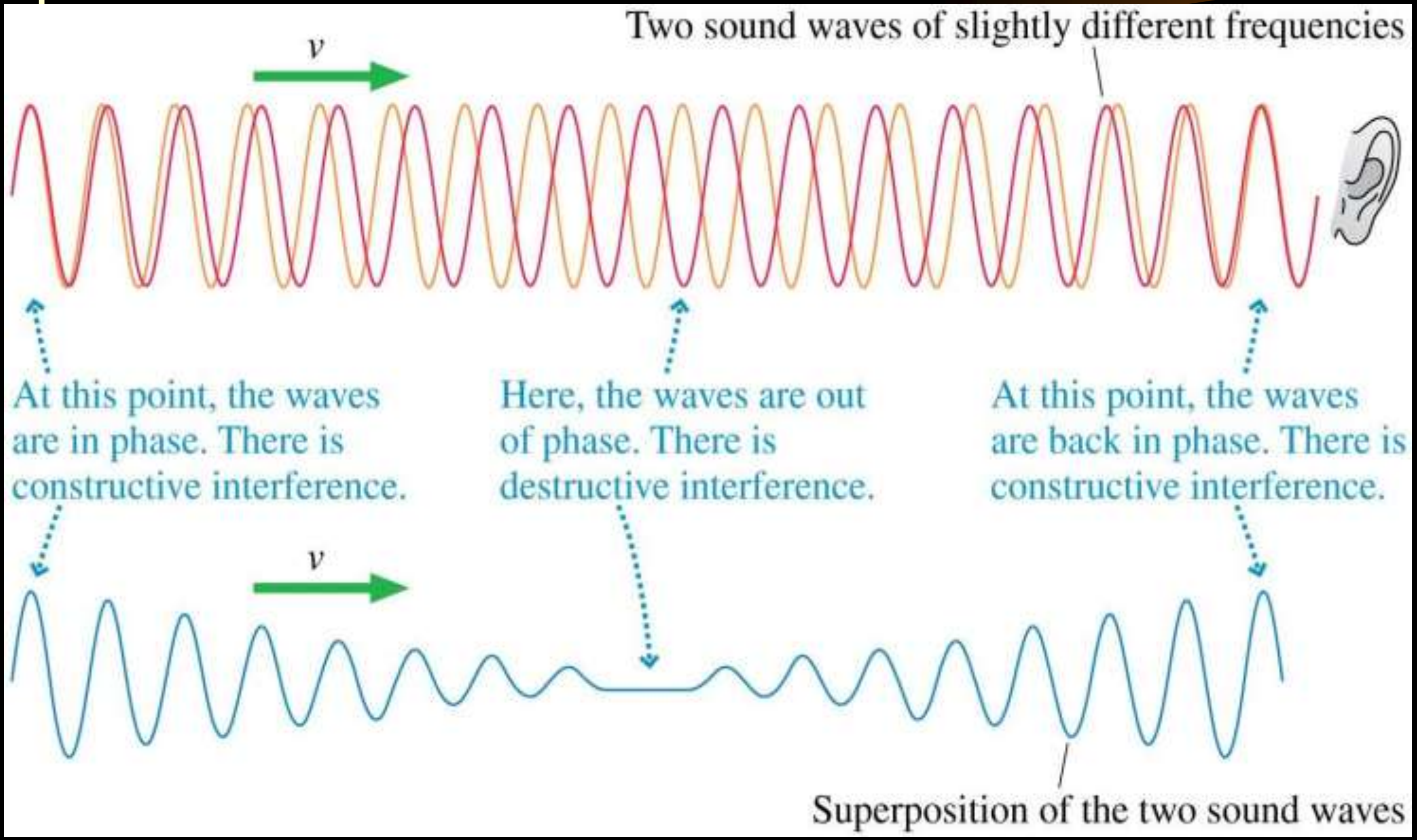
Interference Practice

- If waves A and B were aligned on top of each other, what would the resultant wave look like?



Beats

- The superposition of two waves with slightly different frequencies can create a wave whose amplitude shows a periodic variation.
- The ear hears a single tone that is *modulated*. The distinctive sound pattern is called **beats**.



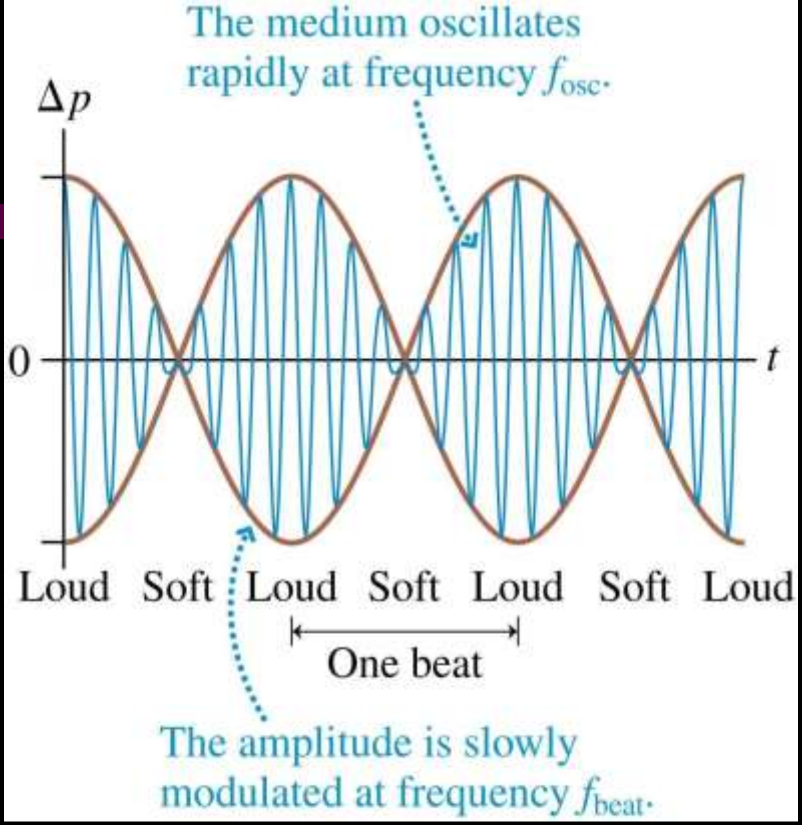
Beats

- The air oscillates against your eardrum at frequency

$$f_{\text{osc}} = \frac{1}{2}(f_1 + f_2)$$

- The *beat frequency* is the *difference* between two frequencies that differ slightly:

$$f_{\text{beat}} = |f_1 - f_2|$$



f_{osc} determines the pitch, f_{beat} determines the frequency of the modulations.

Standing waves on a string

- Waves traveling in opposite directions on a taut string interfere with each other.
- The result is a *standing wave* pattern that does not move on the string.
- *Destructive interference* occurs where the wave displacements cancel, and *constructive interference* occurs where the displacements add.
- At the *nodes* no motion occurs, and at the *antinodes* the amplitude of the motion is greatest.

Standing Waves

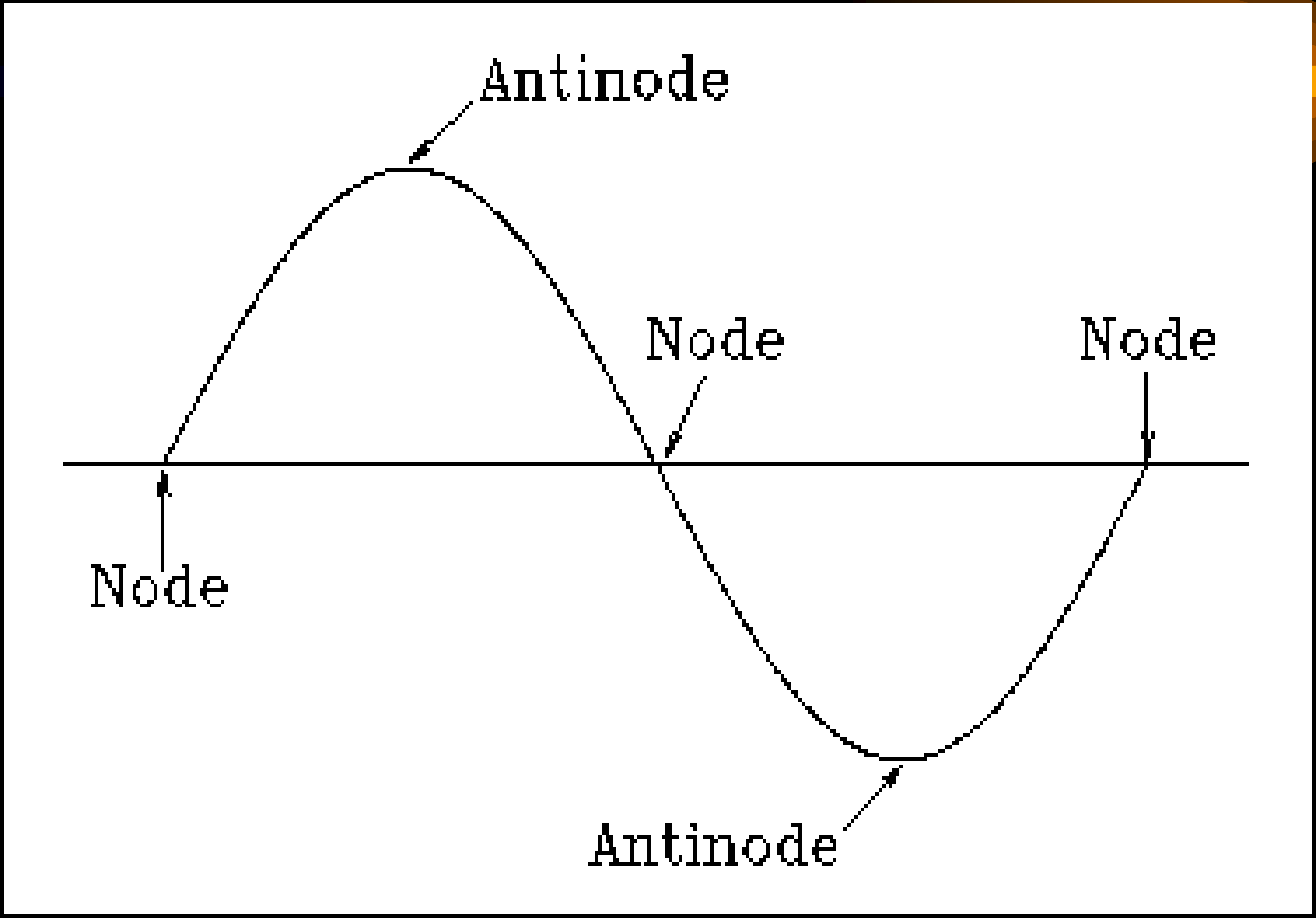
- When a wave appears to just move up and down instead of transmitting energy side to side it is called a standing wave.

- These waves appear to be standing still, hence “Standing Wave”.

Nodes and Antinodes

- Nodes and Antinodes are used to help describe and understand waves.
- A “node” is where waves meet and cause zero displacement of the medium.
  - No displacement
- An “antinode” is where waves meet and cause the largest displacement of the medium.
  - Anti no displacement

Nodes and Antinodes

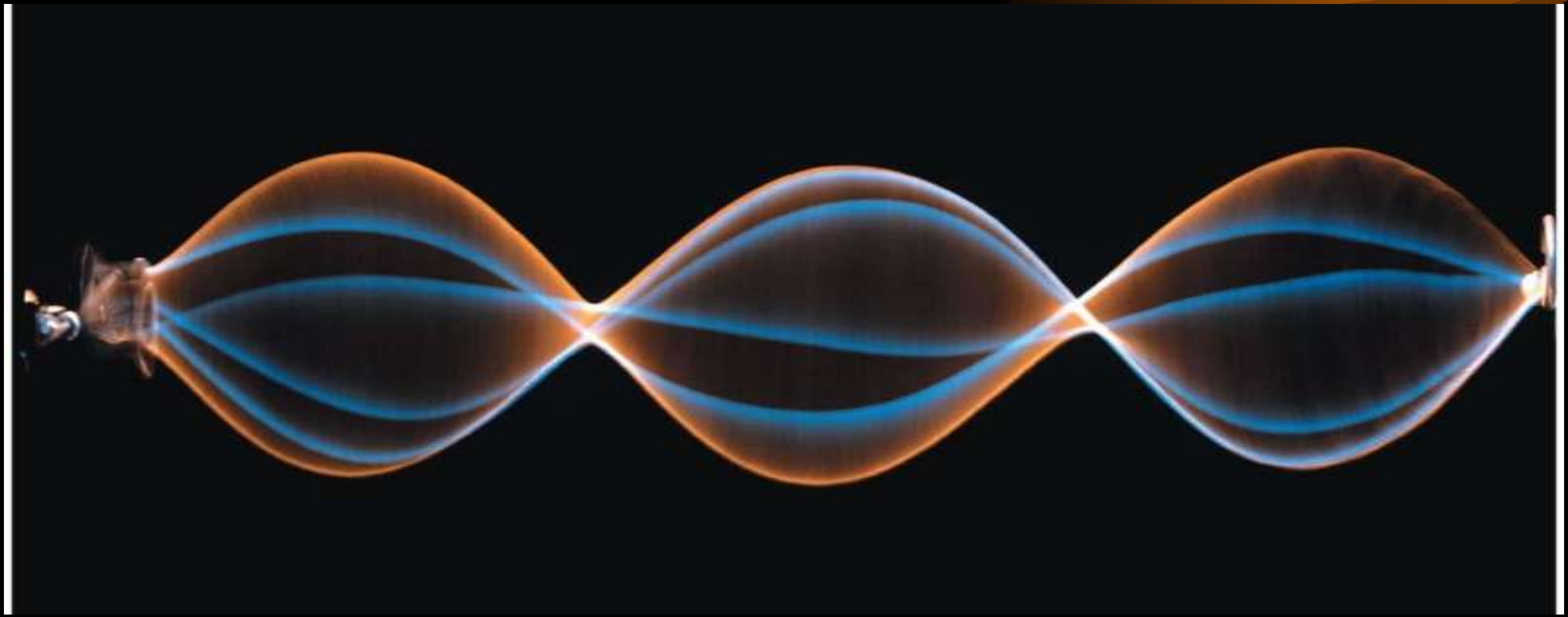


Standing Waves

- Waves that are "trapped" and cannot travel in either direction are called *standing waves*.
- **Individual points on a string oscillate up and down, but the wave itself does not travel.**
- It is called a **standing wave** because the crests and troughs "stand in place" as it oscillates.

Standing Waves

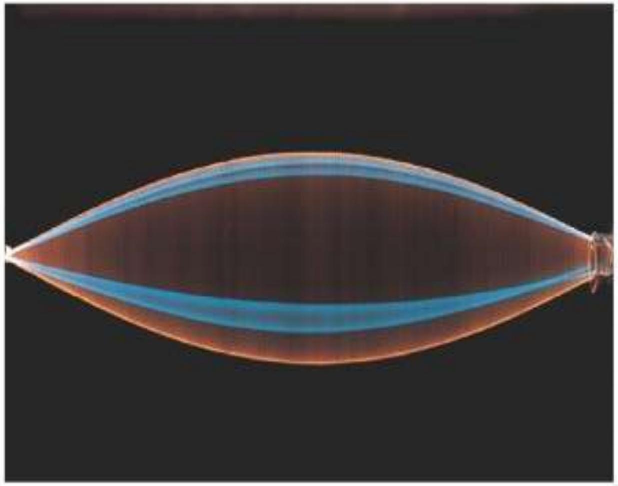
- The superposition of waves on a string can lead to a wave that oscillates in place—a **standing wave**.



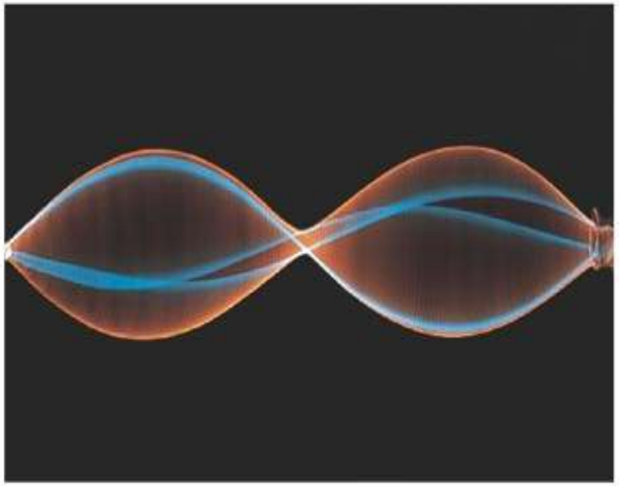
- You'll learn the patterns of standing waves on strings and standing sound waves in tubes.

Standing waves on a stretched string.

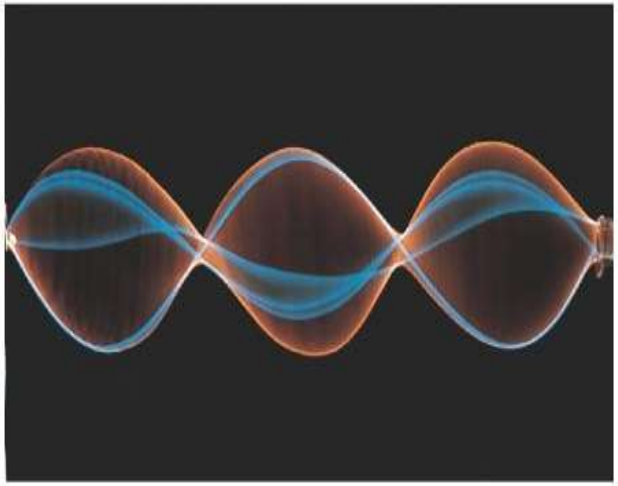
(a) String is one-half wavelength long.



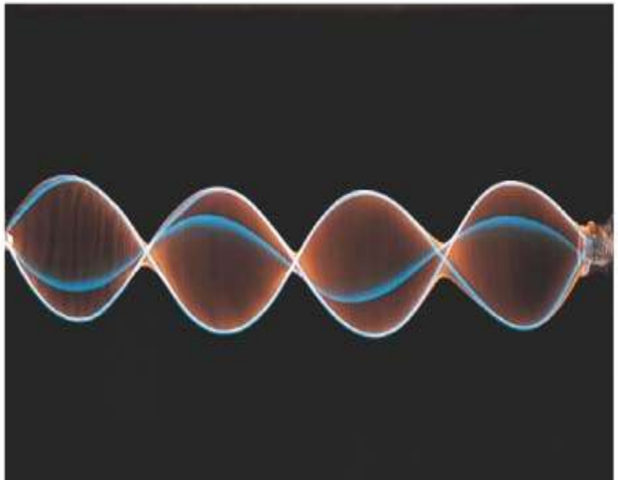
(b) String is one wavelength long.



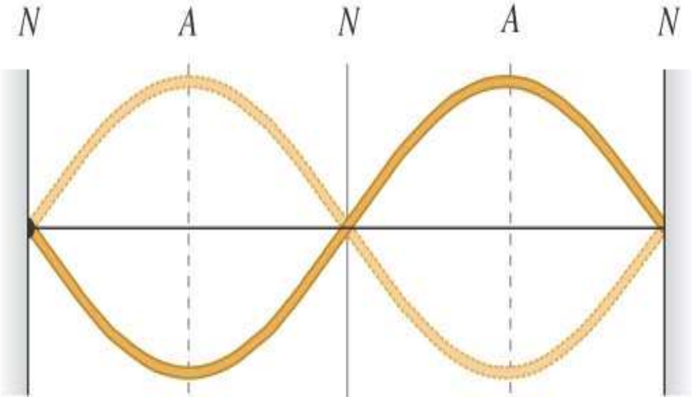
(c) String is one and a half wavelengths long.



(d) String is two wavelengths long.



(e) The shape of the string in (b) at two different instants



N = **nodes:** points at which the string never moves

A = **antinodes:** points at which the amplitude of string motion is greatest

The Fundamental and Higher Harmonics

- The first mode of the standing-wave modes has the frequency

$$f_1 = \frac{v}{2L}$$

- This frequency is the **fundamental frequency** of the string.

The Fundamental and Higher Harmonics

- The frequency in terms of the fundamental frequency is

$$f_m = mf_1 \qquad\qquad m = 1, 2, 3, 4, \ldots$$

- **The allowed standing-wave frequencies are all integer multiples of the fundamental frequency.**

- The sequence of possible frequencies is called a set of **harmonics.**

- Frequencies above the fundamental frequency are referred to as **higher harmonics.**

Normal modes of a string

- For a taut string fixed at both ends, the possible wavelengths are $\lambda_n = 2L/n$ and the possible frequencies are $f_n = n\,v/2L = nf_1$, where $n = 1, 2, 3, \ldots$
- f_1 is the *fundamental frequency*, f_2 is the second harmonic (first overtone), f_3 is the third harmonic (second overtone), etc.
- Figure illustrates the first four harmonics.